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PRACTICAL 6 OF PYTHON FOR DS

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
import pandas as pd
df = pd.read_csv("Students.csv")
print(df.head())
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80
4	105	Rohan	BSc IT	55	70

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0	101	Amit	BSc IT	78	85
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6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape)
print(df.columns)
print(df.dtypes)
print(df.describe())
```

```
(5, 5)
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

	Roll No	Marks	Attendance
count	5.000000	5.000000	5.000000
mean	103.000000	71.600000	83.400000
std	1.581139	12.54193	8.820431
min	101.000000	55.000000	70.000000
25%	102.000000	65.000000	80.000000
50%	103.000000	72.000000	85.000000
75%	104.000000	78.000000	90.000000
max	105.000000	88.000000	92.000000

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60

```
print(df[df["Marks"] > 60])
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks

```
print(df.sort_values("Marks", ascending=False).head())
```

	Roll No	Name	Department	Marks	Attendance
2	103	Rahul	BSc IT	88	92
0	101	Amit	BSc IT	78	85
3	104	Neha	BSc CS	72	80
1	102	Priya	BSc CS	65	90
4	105	Rohan	BSc IT	55	70

```
print(df.sort_values("Marks").head())
```

	Roll No	Name	Department	Marks	Attendance
4	105	Rohan	BSc IT	55	70
1	102	Priya	BSc CS	65	90
3	104	Neha	BSc CS	72	80
0	101	Amit	BSc IT	78	85
2	103	Rahul	BSc IT	88	92

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print(df["Marks"].max())  
print(df["Marks"].min())  
print(df["Marks"].mean())  
print(df["Marks"].sum())
```

```
88  
55  
71.6  
358
```